TASK 2 : Sales Performance Analysis

A. Import Libraries

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.style.use('ggplot')
```

B. Load the Dataset

In [6]:
```
import pandas as pd

df = pd.read_csv("superstore_final_dataset (1).csv", encoding='latin1')
df.head()
```

Out[6]:

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Nam
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gu
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gu
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van Hu
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donn
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donn

5 rows × 21 columns

c. Understanding Data

In [8]:
```
df.info()
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row_ID         9800 non-null   int64
 1   Order_ID       9800 non-null   str
 2   Order_Date     9800 non-null   str
 3   Ship_Date      9800 non-null   str
 4   Ship_Mode      9800 non-null   str
 5   Customer_ID    9800 non-null   str
 6   Customer_Name  9800 non-null   str
 7   Segment        9800 non-null   str
 8   Country        9800 non-null   str
 9   City           9800 non-null   str
 10  State          9800 non-null   str
 11  Postal_Code    9789 non-null   float64
 12  Region         9800 non-null   str
 13  Product_ID     9800 non-null   str
 14  Category       9800 non-null   str
 15  Sub_Category   9800 non-null   str
 16  Product_Name   9800 non-null   str
 17  Sales          9800 non-null   float64
dtypes: float64(2), int64(1), str(15)
memory usage: 1.3 MB
```

Out[8]:

	Row_ID	Postal_Code	Sales
count	9800.000000	9789.000000	9800.000000
mean	4900.500000	55273.322403	230.769059
std	2829.160653	32041.223413	626.651875
min	1.000000	1040.000000	0.444000
25%	2450.750000	23223.000000	17.248000
50%	4900.500000	58103.000000	54.490000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

D. Cleaning Dataset

In [9]:
```
df = df.drop(columns=['Unnamed: 18','Unnamed: 19','Unnamed: 20'], errors='ignore
```

In [10]:
```
df.columns
```

Out[10]:
```
Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
       'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
       'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
       'Product_Name', 'Sales'],
      dtype='str')
```

E. Converting Columns to Correct Type

In [13]:
```
df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True)
df['Ship_Date'] = pd.to_datetime(df['Ship_Date'], dayfirst=True)
```

In [14]:
```
df['Sales'] = pd.to_numeric(df['Sales'], errors='coerce')
```

F. Handling Missing Values

In [15]:
```
df.isnull().sum()
```

Out[15]:
```
Row_ID            0
Order_ID          0
Order_Date        0
Ship_Date         0
Ship_Mode         0
Customer_ID       0
Customer_Name     0
Segment           0
Country           0
City              0
State             0
Postal_Code      11
Region            0
Product_ID        0
Category          0
Sub_Category      0
Product_Name      0
Sales             0
dtype: int64
```

In [16]:
```
df = df.dropna(subset=['Sales'])
```

G. Creating Time Columns

In [17]:
```
df['Year'] = df['Order_Date'].dt.year
df['Month'] = df['Order_Date'].dt.month
df['Month_Name'] = df['Order_Date'].dt.month_name()
```

H. Creating KPIs

1. Total Revenue

In [18]:
```
total_revenue = df['Sales'].sum()
print("Total Revenue:", total_revenue)
```

```
Total Revenue: 2261536.7827
```

2. Average Order Value

In [19]:
```
total_orders = df['Order_ID'].nunique()
aov = total_revenue / total_orders
print("Average Order Value:", aov)
```

```
Average Order Value: 459.4751691791954
```

3. Total Order

In [20]:
```
print("Total Orders:", total_orders)
```

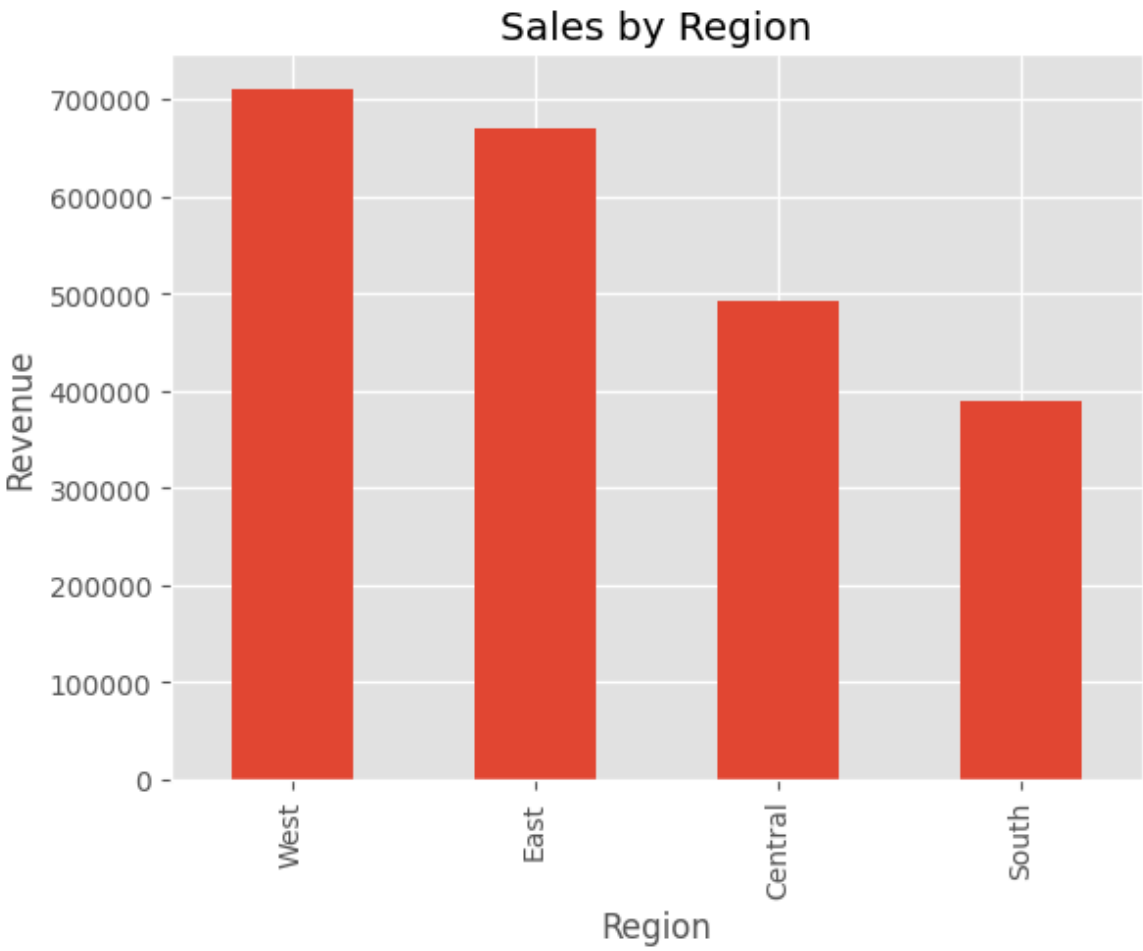
```
Total Orders: 4922
```

I. Regional Sales Analysis

In [24]:
```
region_sales = df.groupby('Region')['Sales'].sum().sort_values(ascending=False)
region_sales
```

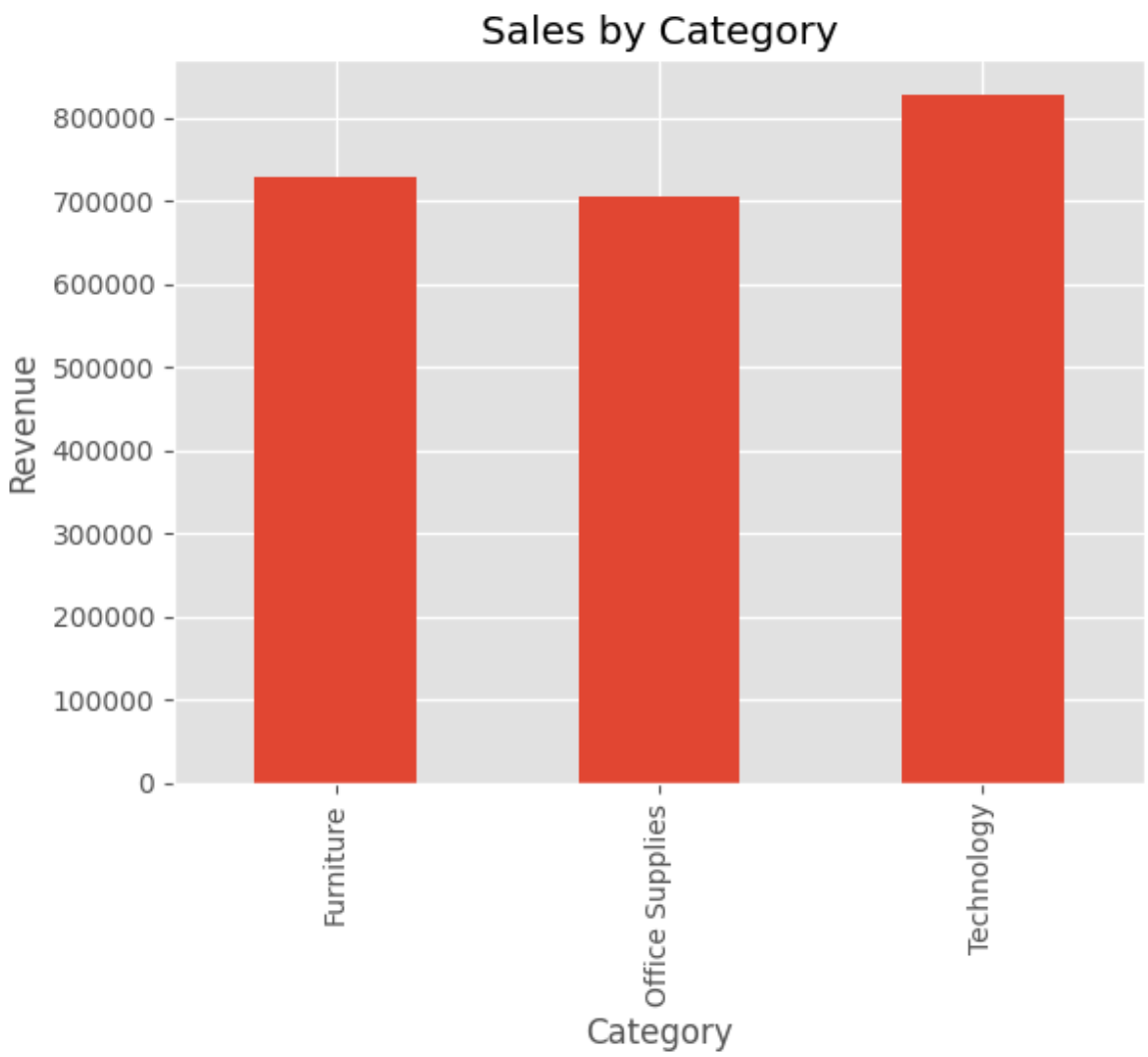
Out[24]:
```
Region
West       710219.6845
East       669518.7260
Central    492646.9132
South      389151.4590
Name: Sales, dtype: float64
```

In [25]:
```
region_sales.plot(kind='bar')
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Revenue")
plt.show()
```



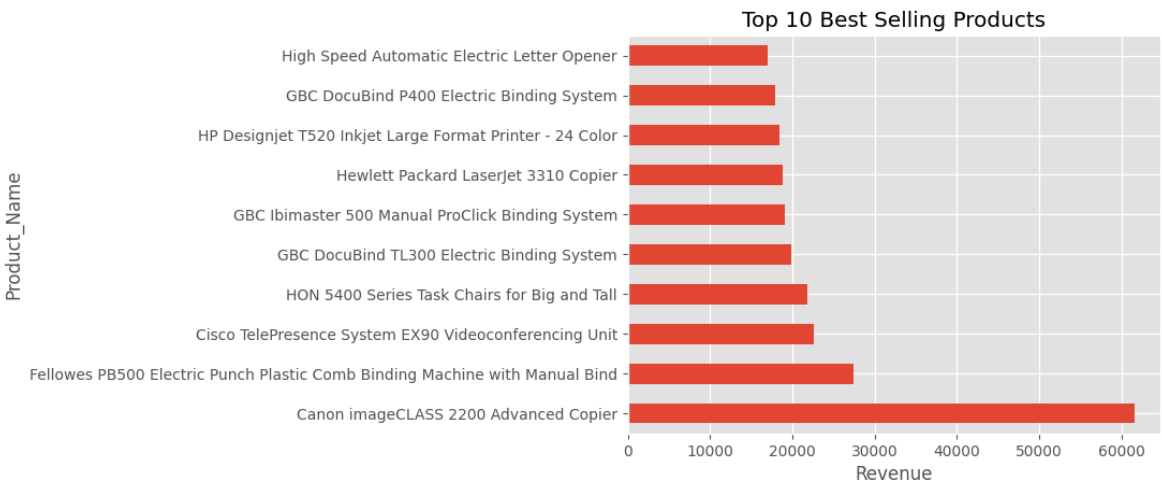
J. Sales by Category

In [26]:
```
category_sales = df.groupby('Category')['Sales'].sum()
category_sales.plot(kind='bar')
plt.title("Sales by Category")
plt.ylabel("Revenue")
plt.show()
```



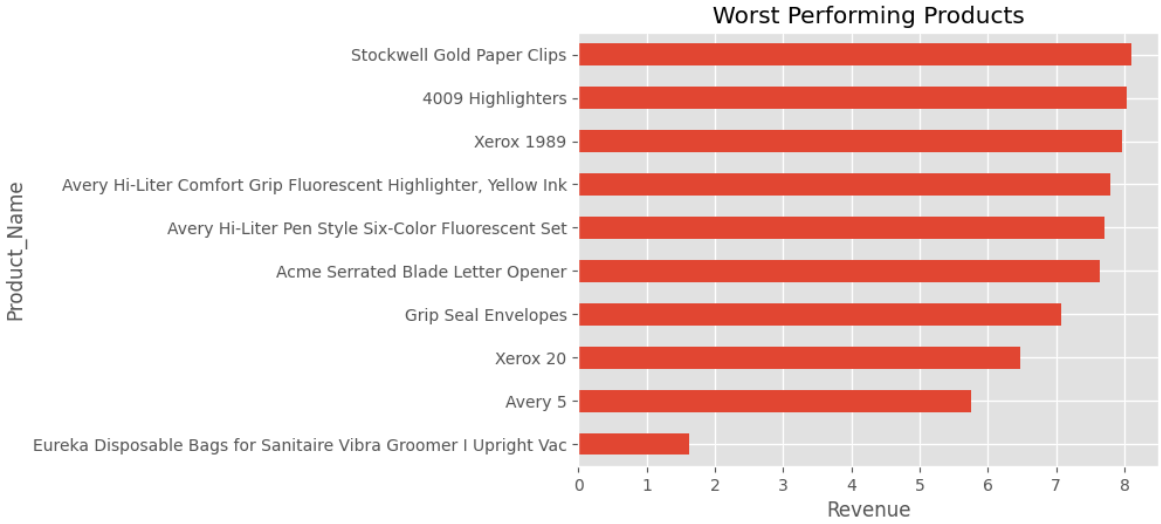
K. Top 10 Best Selling Products

In [28]:
```
top_products = df.groupby('Product_Name')['Sales'].sum().sort_values(ascending=F
top_products.plot(kind='barh')
plt.title("Top 10 Best Selling Products")
plt.xlabel("Revenue")
plt.show()
```



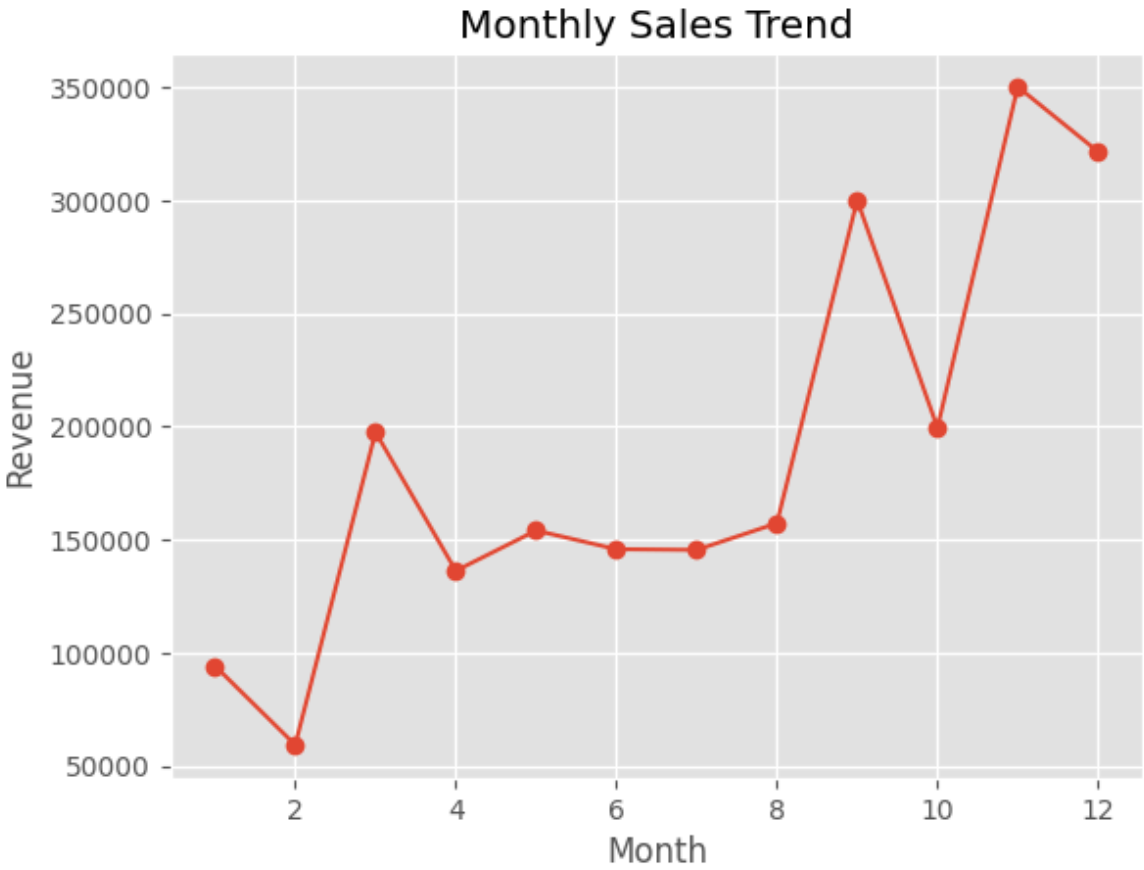
L. Worst Performing Proudcts

In [30]:
```
worst_products = df.groupby('Product_Name')['Sales'].sum().sort_values().head(10
worst_products.plot(kind='barh')
plt.title("Worst Performing Products")
plt.xlabel("Revenue")
plt.show()
```



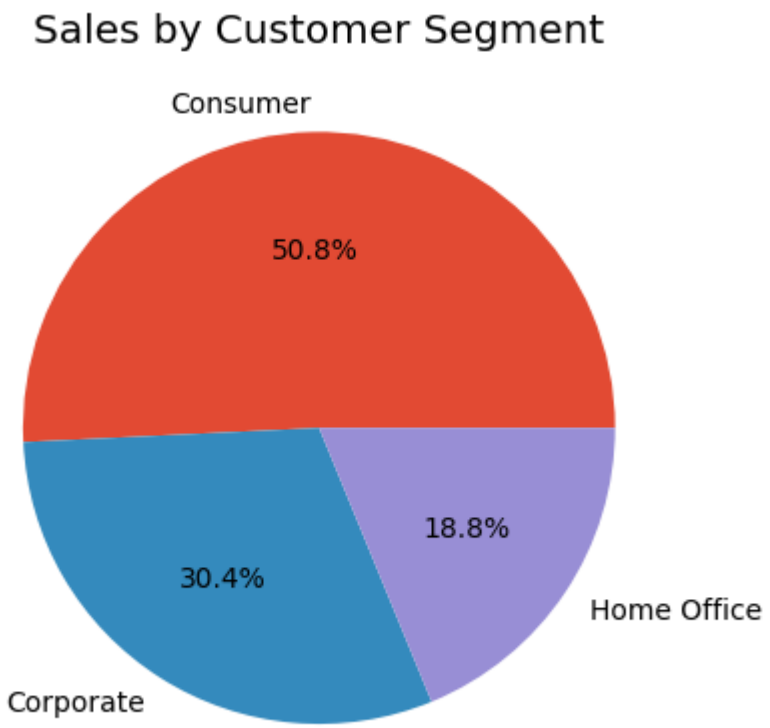
M. Monthly Sales Trend

In [31]:
```
monthly_sales = df.groupby('Month')['Sales'].sum()
monthly_sales.plot(kind='line', marker='o')
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.show()
```



N. Sales By Customer Segment

In [32]:
```
segment_sales = df.groupby('Segment')['Sales'].sum()
segment_sales.plot(kind='pie', autopct='%1.1f%%')
plt.title("Sales by Customer Segment")
plt.show()
```



O. Key Business Insights

1. West region generates the highest revenue compared to other regions.
2. Technology category contributes the most to total sales.
3. A small number of products generate the majority of revenue.
4. Sales increase significantly during the last quarter of the year.
5. Consumer segment contributes the largest portion of revenue.

P. 5 Suggested Tactical Improvements

1. Focus marketing campaigns in high-performing regions to maximize revenue.
2. Increase inventory and promotion of top-selling products.
3. Improve marketing strategies for low-performing products.
4. Use seasonal sales patterns to launch discount campaigns during peak months.
5. Target the consumer segment with personalized offers and loyalty programs